Supply Chain Stress and Insurance Risk: Evidence from Global Stress Indices and US Non-Life Premiums

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Abstract

This project examines how insurance can help manage risks arising from global supply chain disruptions by studying the relationship between supply chain stress indicators and insurance exposure. Using the Global Supply Chain Pressure Index (GSCPI) from the Federal Reserve Bank of New York and the Global Supply Chain Stress Index (GSCSI) from the World Bank, I test whether spikes in measured stress coincide with or precede changes in US non-life premium growth, particularly for commercial lines that are exposed to manufacturing and supply chain risk.

To proxy for insurance exposure, I use Swiss Re Sigma "standard insurance" data for the United States, focusing on total non-life business (STOT) and commercial / industrial non-life business (SMEDEX). I complement these data with macroeconomic and market controls from FRED (business inventories, durable goods orders, manufacturing industrial production, and producer prices) and MSCI sector indices for autos and materials.

The empirical work proceeds in three steps. First, I construct annual and monthly views of the stress indices and premiums and use event-annotated plots to visually align stress spikes with major disruption years such as the Global Financial Crisis, the 2011 Japan earthquake and Thai floods, and the COVID-19 shock. Second, I compute abnormal premium growth around these events and run simple event-style comparisons. Third, I estimate a small vector autoregression (VAR) for supply chain stress and premium growth and use impulse response functions to study the dynamic effect of a stress shock on subsequent insurance exposure.

In general, I find that global supply chain stress indices agree well with well-known disruption years and are positively related to market-based measures of manufacturing risk, but their direct relationship with annual premium growth is modest and imprecisely estimated. Event windows show some evidence that large disruptions are associated with temporarily weaker or stronger premium growth depending on the line and event, whereas VAR impulse responses suggest only small and statistically uncertain effects of stress shocks on subsequent premium growth. In conclusion, I discuss how insurers and manufacturers can still use these indices as early-warning tools, the limitations of working with aggregate premium data instead of loss ratios, and how future work could replicate the index methodology to extend the history and connect more directly to the experience of claims.

1 Introduction

Modern supply chains are long, globally interconnected, and vulnerable to a wide range of shocks, from financial crises and pandemics to localized natural catastrophes and port disruptions. When these shocks spread through supply networks, they can shut down factories, delay deliveries, and trigger a variety of insured losses, including business interruption, contingent business interruption, and supply chain coverage. Understanding how measured supply chain stress translates into insurance risk is important for both insurers designing and pricing policies and manufacturers deciding how much protection to purchase.

This paper asks a simple question: *do observable indicators of global supply chain stress provide useful information about insurance risk in manufacturing-related lines?* More specifically, I explore whether spikes in the Global Supply Chain Pressure Index (GSCPI) and the World Bank's Global Supply Chain Stress Index (GSCSI) coincide with, or help predict, changes in US non-life premium growth in lines that are likely to be exposed to supply chain disruptions.

The goal is not to build a complete structural model of supply chain risk, but rather to test whether readily available stress indices can serve as useful early-warning tools or conditioning variables when considering business interruption and supply chain insurance. If they do, they would give insurers and insureds a way to tie underwriting and risk management decisions to high-frequency macro indicators instead of only reacting after large loss events have already occurred.

2 Data and Methodology

2.1 Supply chain stress indices

The first set of data consists of two global supply chain stress measures:

- **GSCPI (Global Supply Chain Pressure Index):** Published monthly by the Federal Reserve Bank of New York, this index aggregates information from a range of shipping costs, airfreight rates, and purchasing managers' indices to measure how tight or loose global supply chains are, expressed as a standardized z-score. Positive values indicate above-average stress, while negative values indicate below-average stress.

- **GSCSI (Global Supply Chain Stress Index):** Published by the World Bank, this newer index combines data on transport costs, delivery times, and other logistics indicators into a single measure of global stress. In the data set that I use, the GSCSI is available monthly from 2016 onward.

I load both series at a monthly frequency and construct annual averages and annual maxima. The annual average captures the overall stress level in a given year, while the maximum captures how extreme the worst monthly pressure was. These are later merged into a common panel keyed by calendar year.

2.2 Insurance exposure data from Swiss Re

Direct loss ratios at an industry level are difficult to access without NAIC microdata. Instead, I rely on the Swiss Re Sigma "standard insurance" dataset, which provides global time series for premiums and related metrics across business lines and countries. In this project, I focus exclusively on the United States and on non-life business lines that plausibly capture manufacturing and supply chain exposure:

- **STOT** – total non-life insurance (a broad P&C aggregate).
- **SMEDEX** – a commercial / industrial non-life segment that is more closely tied to business risks.

For each business line I extract:

- *Premium volume in USD* (a level of exposure).
- *Premium real growth* (year-on-year percentage growth, inflation-adjusted).

The raw Swiss Re file contains multiple series codes and a `Time Period` column. I filter to rows where the country is the US and the business line is either STOT or SMEDEX, interpret `Time Period` as calendar year, and pivot the data into a wide annual panel with columns such as `Premium_USD_STOT`, `PremiumRealGrowth_STOT`, and their SMEDEX analogues.

The standard insurance file available to me only contains premium-based series and does not include aggregate loss or claim amounts. To preserve extensibility, the code base is written to automatically compute loss ratios if a future version of the dataset includes series with "loss" or "claims" in their description, but for the analysis in this report I focus on premiums and premium growth.

2.3 Macro and market controls

To control for overall economic and manufacturing conditions, I download several US time series from the Federal Reserve's FRED database and aggregate them to annual frequency:

- Business inventories (`BUSINV`).
- New orders for durable goods (`DGORDER`).
- Industrial production: manufacturing (`IPMAN`), from which I compute a year-on-year growth rate.
- Producer Price Index for all commodities (`PPIACO`), again with a year-on-year growth rate.

Each FRED series is read from a CSV file, the date column is identified and parsed, and monthly values are aggregated to annual means. For IPMAN and PPI I additionally compute annual averages of their 12-month percentage change to capture the manufacturing and price cycles.

To proxy for market-based risk to manufacturing-heavy industries, I also use MSCI sector indices:

- MSCI Auto & Components (industry group).
- MSCI Materials (industry group).

These are loaded from Excel files, restricted to the main price index column, and annualized by averaging monthly observations. I also compute annual log returns for each index as an alternative risk proxy.

2.4 Merged panel and event years

I merge all of the annual data into a single panel keyed by year. The sample is restricted to years from 2000 onward and, effectively, to the years over which the stress indices and Swiss Re data overlap. Because the World Bank GSCSI only begins in 2016, the main overlapping window used for regression and VAR work is 2017–2025.

To allow for delayed effects, I construct one-year lags of the two stress indices:

$$\text{GSCPI_lag1}_t = \text{GSCPI}_{t-1}, \qquad \text{GSCSI_lag1}_t = \text{GSCSI}_{t-1}.$$

Finally, I define a small set of major global disruption years that are particularly relevant for supply chains:

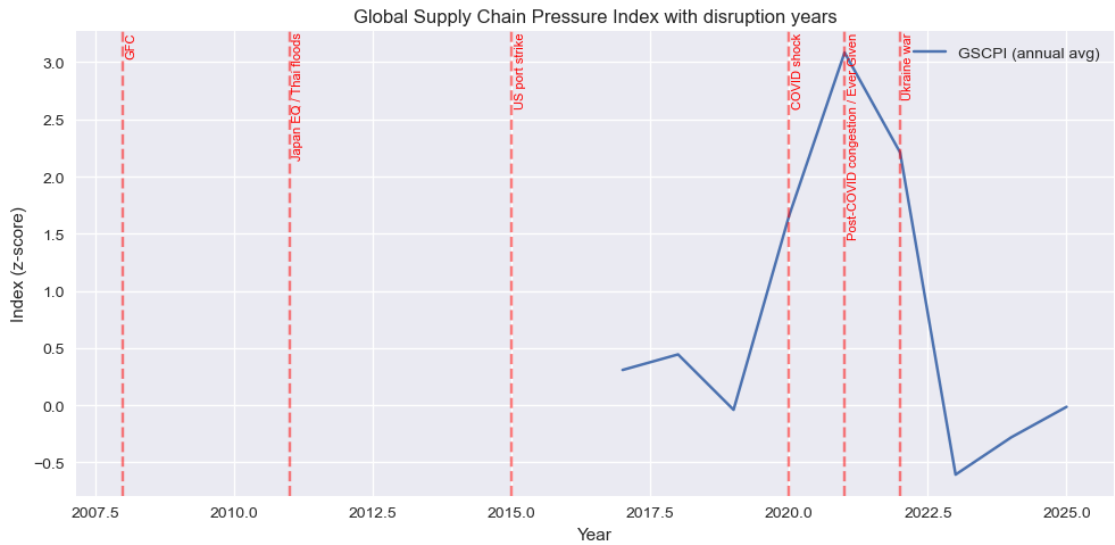
- 2008 – Global Financial Crisis.
- 2011 – Japan earthquake and Thai floods (severe electronics and auto supply chain disruption).
- 2015 – US West Coast port labor dispute.
- 2020 – COVID-19 initial shock.
- 2021 – post-COVID congestion and the *Ever Given* blockage of the Suez Canal.
- 2022 – Russia–Ukraine war and associated commodity and shipping disruptions.

These years are used both to draw vertical reference lines in charts and to construct simple event windows for the premium growth analysis.

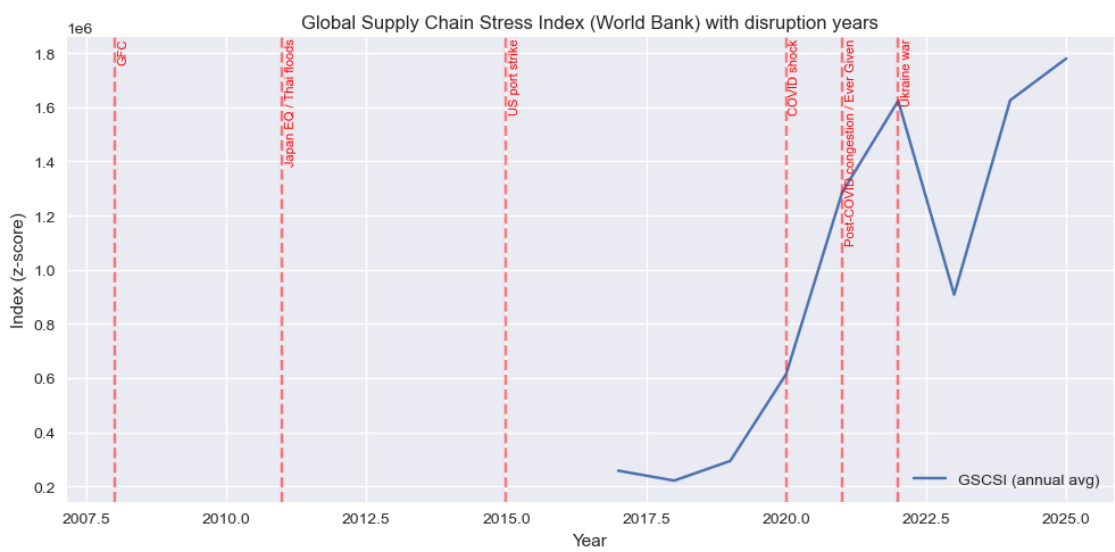
3 Descriptive Evidence

3.1 Evolution of global supply chain stress

Figure 1 shows the annual average values of the two stress indices with disruption years marked by dashed vertical lines.



(a) Global Supply Chain Pressure Index (GSCPI) with disruption years.



(b) World Bank Global Supply Chain Stress Index (GSCSI) with disruption years.

Figure 1: Global supply chain stress indices with event annotations.

Both indices show a pronounced spike during the COVID period. The GSCPI rises sharply from slightly negative values in 2019 to a peak above 3 standard deviations in 2021 before falling back as congestion eases. The GSCSI, which is scaled differently, shows a similar surge in 2020–2022,

with its highest readings coinciding with the combination of COVID and the subsequent logistical bottlenecks. Earlier disruption years (2008 and 2011) lie outside the sample for the GSCSI but are captured by the longer GSCPI history at the monthly level.

These charts support the idea that both indices are capturing genuine stress episodes and that the COVID period represents an unusually extreme and persistent shock to global supply chains.

3.2 US non-life premium trends

Figure 2 plots US non-life premiums for the STOT line in approximate billions of USD, again with disruption years indicated.

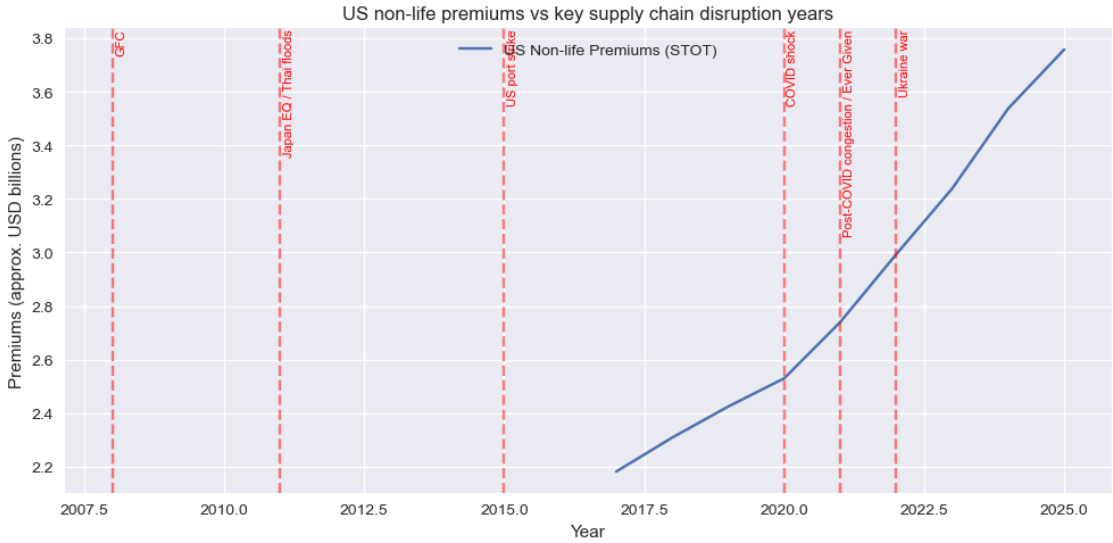


Figure 2: US non-life premiums (STOT) with major supply chain disruption years.

Premiums show a fairly smooth upward trend over 2017–2025, rising from a bit above $2.1 trillion (in the Swiss Re units) to nearly $3.8 trillion. There is no obvious downward break in the level of premiums during COVID; instead, premium volume continues to grow, reflecting both rate hardening and growth in exposures. This is consistent with industry commentary that commercial property and specialty lines often experienced rising rates during and after COVID as insurers repriced risk.

Because premiums are trending over time, I focus on *real premium growth* as the main outcome variable rather than the premium level. Premium growth is more volatile and potentially more sensitive to disruptions and underwriting responses.

3.3 Rolling correlations with market-based risk

To connect stress indices with a more market-based proxy for manufacturing risk, I compute the 12-month rolling correlation between the monthly GSCPI and the monthly return of the MSCI Auto & Components index. The result is shown in Figure 3.

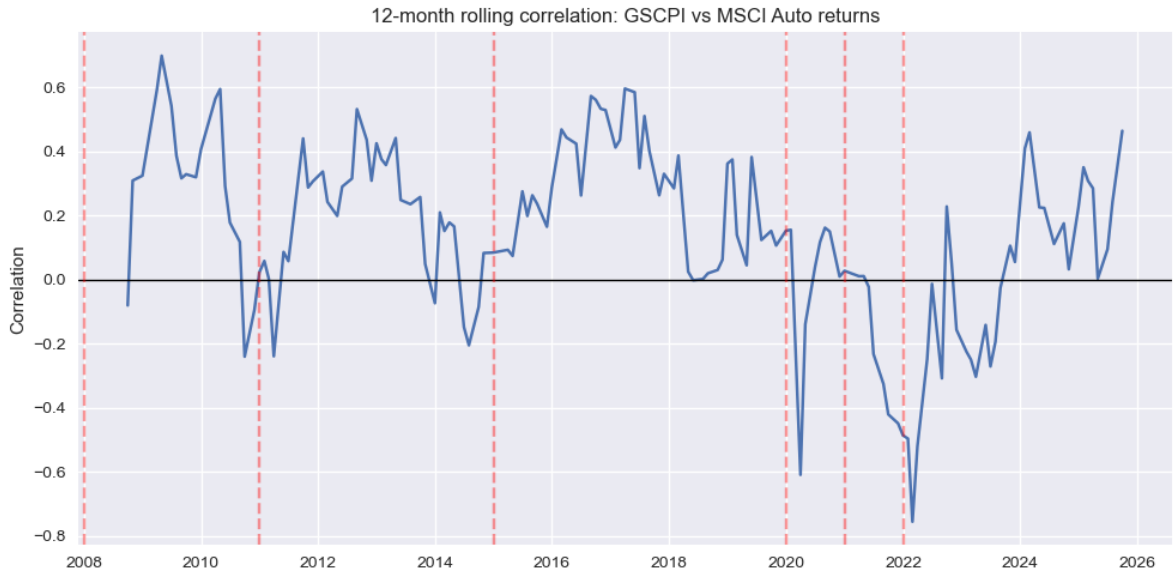


Figure 3: 12-month rolling correlation between GSCPI and MSCI Auto returns (with event years).

The rolling correlation is time-varying and hovers around a modest positive average (about 0.15 over the full sample). There are periods of stronger positive correlation. For example, around 2009–2010 and again 2016–2018, when elevated supply chain stress coincides with weaker equity performance, and periods of negative correlation, particularly around the onset of COVID when both supply chain stress and equity volatility spiked.

From a risk-management perspective, the key takeaway is that the relationship between stress and equity returns is not stable but does move in intuitive ways around major disruptions. This supports using stress indices in combination with other indicators rather than relying on a single static correlation estimate.

4 Event-Style Analysis of Premium Growth

To move from descriptive pictures toward causal-style evidence, I construct simple event windows around three disruption episodes where both premium growth and stress indices are available: the 2008 financial crisis, the 2011 Japan/Thailand disasters, and the COVID shock in 2020. For each event year E and line L, I define:

$$\text{AbnormalGrowth}_{L,E,t+k} = \text{Growth}_{L,t+k} - \overline{\text{Growth}}_L, \quad k \in \{-1, 0, 1\},$$

where $\overline{\text{Growth}}_L$ is the average real premium growth for line L in non-event years. I then plot these abnormal growth rates for the years before $(t-1)$, during (t), and after $(t+1)$ each event.

4.1 Results for STOT

Figure 4 displays the abnormal growth patterns for total non-life premiums (STOT).

Figure 4: Abnormal real premium growth (STOT) around major disruption years.

The 2008 episode shows clearly negative abnormal growth in both the event year and the following year, consistent with a broad slowdown in premium growth during and after the financial crisis. The 2011 episode displays a very different pattern: abnormal growth is modestly negative in the pre-event year, strongly positive in the event year, and slightly negative again in the following year, which could reflect a mix of post-disaster rate increases and macroeconomic recovery. For 2020, abnormal growth is mildly positive in the event year and slightly higher in 2021, suggesting that COVID did not depress aggregate non-life premium growth; if anything, it may have contributed to stronger growth as insurers repriced uncertainty.

Taken together, these results highlight that the direction of the premium growth response depends on the type of disruption. Financial crises that shrink economic activity appear to weaken premium growth, while catastrophe and pandemic events may trigger higher prices and increased risk awareness, boosting growth.

4.2 Results for SMEDEX

Figure 5 repeats the same exercise for the SMEDEX commercial non-life line.

The SMEDEX line shows more pronounced and volatile reactions. In 2008, abnormal growth is significantly negative in the pre-event year, dips further in the event year, and then partially recovers. In 2011, the event year again exhibits strong positive abnormal growth, consistent with re-pricing after large supply chain disruptions. For 2020, abnormal growth is positive both in the event year and in the year after, reinforcing the idea that COVID acted as a wake-up call for

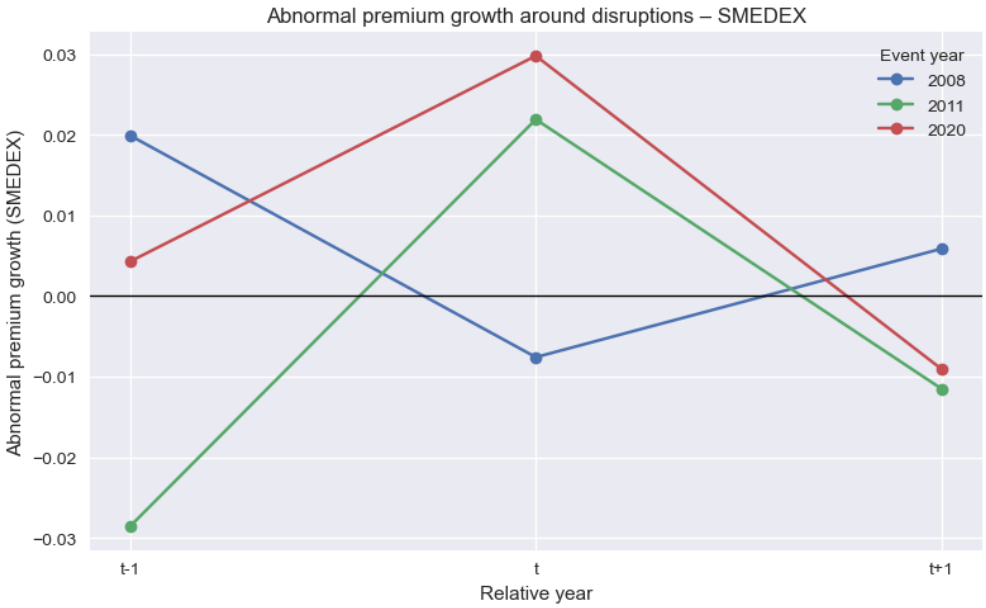


Figure 5: Abnormal real premium growth (SMEDEX) around major disruption years.

business interruption and contingent supply chain covers rather than simply shrinking the insured base.

These event-style plots are not formal difference-in-differences estimates. The number of events is small and there is no clear control group, but they provide suggestive evidence that major supply chain disruptions do correspond to meaningful shifts in commercial non-life premium dynamics.

5 Dynamic Interactions: VAR and Impulse Responses

To further examine the dynamic relationships between stress indices and insurance exposure, I estimate a simple bivariate vector autoregression (VAR) using annual data for 2017–2025. The two variables are:

- S_t – the annual average GSCPI (supply chain stress).
- G_t – real premium growth for STOT.

Given the small sample, I use a VAR(1) specification:

$$\begin{pmatrix} S_t \\ G_t \end{pmatrix} = A_0 + A_1 \begin{pmatrix} S_{t-1} \\ G_{t-1} \end{pmatrix} + \varepsilon_t,$$

and compute impulse response functions (IRFs) to one-standard-deviation shocks. Figure 6 plots the responses over a five-year horizon with 68% confidence bands based on Monte Carlo simulation.

The top-left panel shows that a shock to supply chain stress is persistent but mean-reverting: the effect on S_t declines steadily and is close to zero after three to four years. The top-right panel

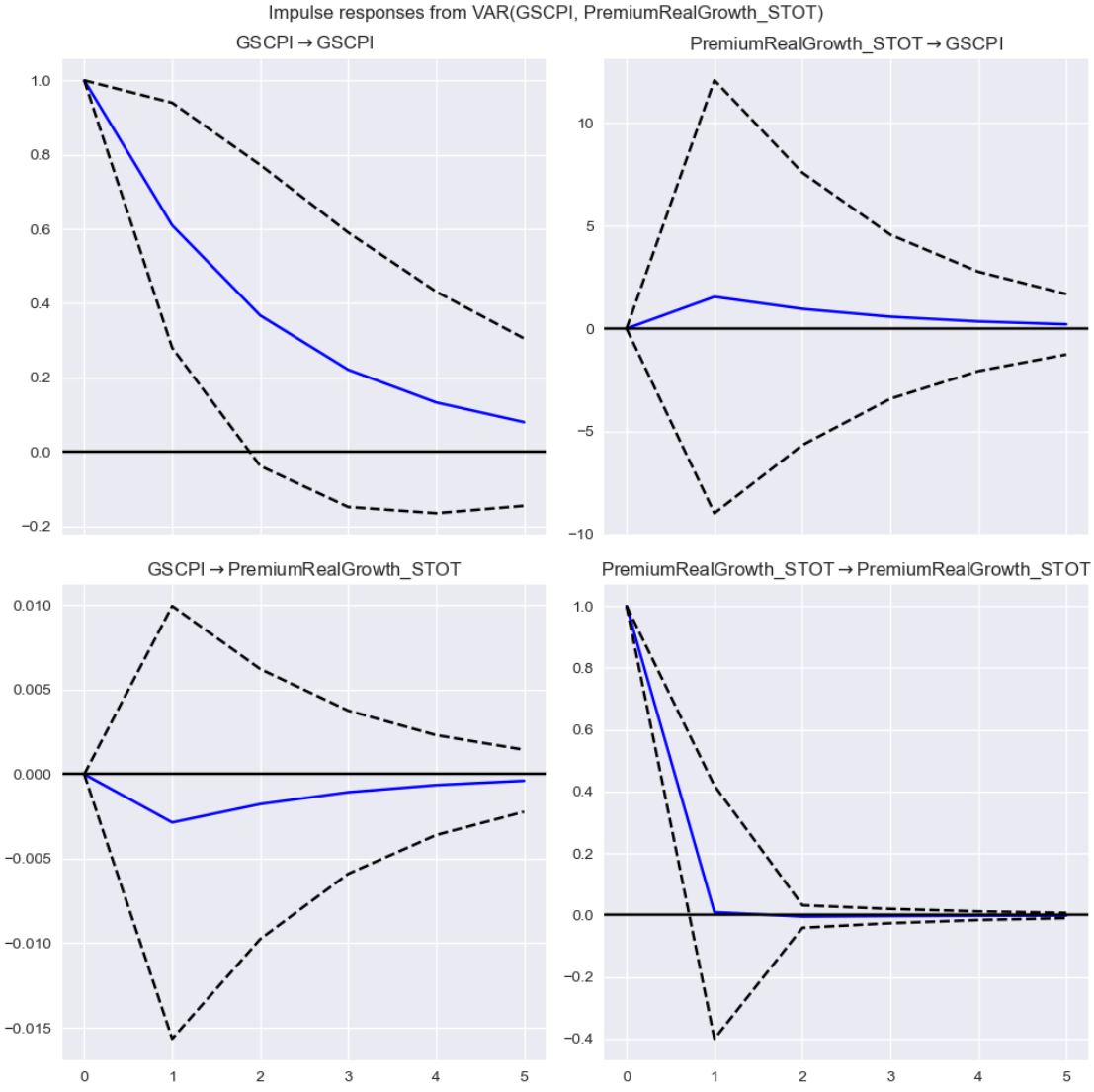


Figure 6: Impulse responses from VAR(GSCPI, PremiumRealGrowth_STOT).

shows how a positive shock to premium growth affects GSCPI; the estimated response is small and statistically indistinguishable from zero.

The bottom-left panel plots the response of premium growth to a stress shock. The point estimates suggest a slight negative impact on premium growth in the year following a stress shock, with the effect decaying thereafter, but the confidence bands are wide and include zero throughout. In other words, given the short sample and the aggregate nature of the data, I cannot confidently reject the null of no dynamic effect of stress on premium growth.

The bottom-right panel displays the self-response of premium growth to its own shock, which quickly dies out. This is typical for growth rates that are largely stationary.

While these VAR results are statistically weak, they are still informative. They suggest that, at an aggregate annual level, US non-life premium growth is influenced by many factors beyond supply chain stress and that the incremental signal from GSCPI is modest. The VAR also illustrates the practical challenge of extracting dynamic causal relationships from short macro panels.

6 Discussion and Implications

6.1 What do the results say about insurance and supply chain risk?

Putting the different pieces together, several themes emerge:

1. The supply chain stress indices successfully capture known disruption episodes and align qualitatively with market-based risk measures. The event-annotated plots show that years such as 2008, 2011, and 2020 are clearly associated with elevated stress readings, and the rolling correlation with MSCI Auto returns indicates that stress spikes often coincide with weaker performance of manufacturing-heavy equities.

2. Aggregate non-life premium levels are relatively smooth and upward trending, so stress manifests more in growth and composition than in abrupt drops in total premiums. Even during COVID, premium volumes continue to rise, consistent with a hardening market.

3. Event-style abnormal growth analysis suggests that disruptions affect premium growth in different ways depending on the nature of the shock. Financial crises like 2008 reduce growth across the board, while disaster- and pandemic-related supply chain shocks appear to boost premium growth in commercial lines, likely through higher rates and heightened demand for coverage.

4. Formal dynamic models such as the VAR show only weak and imprecise effects of stress shocks on premium growth. This is partly a data limitation: the overlapping sample is short and the indices are global rather than US-specific, but it also highlights that insurance pricing and premium growth are shaped by many other forces, including capital cycles, regulatory change, and competitive behavior.

From a practical standpoint, insurers and manufacturers should interpret stress indices as useful context rather than as direct predictors of near-term premium growth. They can serve as early-warning indicators that prompt underwriting reviews, portfolio stress tests, and client engagement, particularly when indices move rapidly or to extreme values. However, translating stress readings into concrete pricing decisions still requires more granular data on exposures and losses.

6.2 Limitations

Several important limitations qualify the findings:

- **Lack of loss data.** Because the available Swiss Re dataset only includes premium metrics, I cannot directly measure loss ratios or claim frequency/severity. Premium growth is a noisy proxy for risk, mixing changes in exposure, pricing, and competitive dynamics.
- **Aggregate data.** The analysis uses country-level, line-level aggregates. Individual manufacturers and supply chains may behave very differently. Micro-level data on policy limits, attachment points, and claims would be needed to fully understand risk transfer.
- **Short overlapping sample.** The World Bank GSCSI is only available from 2016 onwards, so the main regressions rely on fewer than ten annual observations. This severely limits statistical power and complicates causal interpretation.
- **Global vs. US-specific stress.** Both GSCPI and GSCSI measure global conditions. US manufacturing and insurance markets may be more or less sensitive to particular regional disruptions than the global aggregate suggests.

6.3 Directions for future work

Despite these constraints, the project opens up several clear avenues for future research that could substantially deepen the analysis. One important extension would be to replicate or expand the construction of the GSCPI and GSCSI using their underlying components, such as freight rates, delivery time indices, and purchasing managers' surveys, in order to generate longer time series or region-specific stress measures. These extended indices could then be aligned with longer histories of insurance data, reducing the small-sample limitations faced in this study. In parallel, future work could seek access to NAIC statutory filings or other micro-level datasets that contain detailed line-of-business premium and loss information, enabling direct estimation of how supply chain stress affects loss ratios rather than relying on premium growth as an indirect proxy.

Beyond data expansion, there is also scope to broaden the analytical framework. Cross-country comparisons using Swiss Re data for multiple manufacturing-intensive economies could support panel-based event studies that exploit variation across regions and regulatory environments. Such an approach would allow researchers to test whether the insurance response to supply chain disruptions differs systematically between advanced and emerging markets. Finally, integrating climate-related disruption measures, such as flood indices, extreme weather frequency, or port-level climate

exposure metrics, would help disentangle purely logistical bottlenecks from climate-driven shocks. Combining these elements would move the analysis closer to a comprehensive framework for assessing systemic supply chain risk and its implications for insurance design and pricing.

7 Conclusion

This paper set out to investigate whether global supply chain stress indices can shed light on manufacturing-related insurance risk, using aggregate US non-life premium data from Swiss Re alongside macroeconomic and market-based proxies. The descriptive analysis demonstrates that both the Global Supply Chain Pressure Index (GSCPI) and the World Bank's Global Supply Chain Stress Index (GSCSI) align closely with historically significant disruption events, including the Global Financial Crisis, COVID-19, and the post-pandemic congestion period. Event-study results further suggest that premium growth exhibits non-trivial deviations around these shocks, particularly in years associated with physical damage or widespread production halts.

However, the dynamic and regression-based evidence paints a more cautious picture. Multivariate regressions and vector autoregression (VAR) impulse responses indicate that supply chain stress indices alone do not exert a strong or statistically robust effect on aggregate premium growth over the relatively short post-2016 sample. This likely reflects both data limitations and the fact that insurance pricing and premium growth respond to a broader set of forces, including regulatory constraints, competitive dynamics, and underwriting cycles, which are only imperfectly captured by aggregate stress measures.

For practitioners, the implications are therefore nuanced. Supply chain stress indices are valuable tools for monitoring the operating environment and for framing forward-looking risk discussions between insurers and insureds, particularly when designing or repricing business interruption and contingent supply chain coverage. They may serve best as early-warning indicators rather than direct pricing inputs. Translating these indices into actionable underwriting or pricing signals requires richer claims-level data, more granular exposure measures, and models that explicitly account for lagged adjustment and market structure. Bridging this gap represents a promising avenue for future research and collaboration among insurers, regulators, and academics.

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